

Oliver Raczka

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SKILLS

Languages: Python, C, C++, Rust, JavaScript, Java, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, Express, FastAPI, Streamlit, NumPy, Pandas, SciPy, Plotly, SDL2, OpenGL

Tools & Infrastructure: Git, Docker, Docker Compose, GitHub Actions CI/CD, Linux, SQLite, REST APIs, WebSockets

Practices: Test-Driven Development, Unit/Integration Testing (Pytest, Jest), Agile

EXPERIENCE

Freelance Web Developer

May 2024 – Present

- Build and maintain responsive websites for small-business clients using React and Node.js/Express, handling everything from layout and content structure through deployment
- Work directly with clients to scope requirements and turn revision requests into shipped changes, supporting sites after launch as their content and needs change

Desjardins

2024 – Present

Sales Representative

April 2025 – Present

- Consult with clients daily by phone and in person, diagnosing their needs and explaining financial products and trade-offs clearly
- Manage a personal book of client relationships, translating recurring client feedback into process improvements for the team

Event Marketing Intern

January 2024 – January 2025

- Coordinated cross-team event logistics and conducted one-on-one client consultations across multiple concurrent events

PROJECTS

Options Screener | Python, Financial Modelling, REST APIs, LLM Integration

github.com/Ollie1o1/options

- Built and actively maintain a 27-factor options analysis platform (1,125 commits, 159k LOC, 28 subpackages) that ingests live chains from six data providers, computing Black-Scholes pricing, full Greeks, and SVI implied-volatility fits across 8,000+ contracts
- Validated the ranking model against 7 years of end-of-day chains with walk-forward out-of-sample testing, **publishing negative results for strategies that showed no edge**. A forward-cohort gate withholds any model that has not cleared a significance bar
- Traded the screener's signals by hand to **\$9,200+ in realized profit**, choosing entries from its ranked output rather than letting the tool trade on its own
- Engineered a fault-tolerant data layer with multi-tier SQLite caching, retry and rate-limit handling, and per-quote provenance tracking so stale quotes are flagged rather than silently served downstream
- Shipped a Streamlit dashboard, a FastAPI service, and Discord/Telegram bots on a shared core, backed by a 5,000-test Pytest suite and GitHub Actions CI

OVERDRIVE (3D Arena Shooter) | C++, SDL2, OpenGL 3.3, GLSL

github.com/Ollie1o1/3d_shooter

- Built a 3D arena shooter from scratch in C++ with no game engine, running physics on a fixed 60 Hz timestep decoupled from render framerate and interpolating camera motion between steps so movement stays smooth at any FPS
- Implemented Quake-style air-strafting acceleration with swept collision detection, grapple-hook traversal, parry-deflect combat, and 4 enemy types with distinct silhouettes and behaviour driven by a finite state machine across escalating waves
- Wrote a multi-pass OpenGL renderer with instanced enemy drawing, Blinn-Phong point lighting, three-pass bloom, and an optional CRT filter (barrel distortion, chromatic aberration, scanlines), with every texture generated procedurally and no external image assets

NES Emulator | Rust, Systems Programming

github.com/Ollie1o1/better_emulator

- Wrote a cycle-accurate Nintendo Entertainment System emulator implementing the Ricoh 2A03 (6502) instruction set including common unofficial opcodes, a dot-level PPU with Loopy scroll registers, and an APU with all four channels
- Drove the master clock from the CPU with the PPU ticking three times per cycle to match NTSC hardware, propagating NMI and IRQ from the PPU and mapper back to the CPU, and validated the core against the nestest diagnostic ROM
- Implemented three cartridge mappers (NROM, MMC1, UxROM) plus battery-backed saves, covering titles such as Super Mario Bros, Zelda, and Mega Man. Architected the whole system around one owning bus struct instead of reference-counted interior mutability

EDUCATION

University of Guelph

Bachelor of Computing, Major in Computer Science

Guelph, ON

Third Year, Expected April 2028

Relevant Coursework: Operating Systems, Data Structures, Algorithms, Statistics II, Calculus II, Discrete Structures